Reinforcement Learning in Finance: From Current Practice to Standards and Frameworks

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Abstract

This study reviews current practices in applying reinforcement learning (RL) to finance and proposes a road map to advance the field. Using a structured Scopus search and a two-stage screening protocol, we identify 543 records and retain 163 studies (160 empirical, 3 reviews). Our analysis offers four contributions. First, we characterize the state of practice in terms of algorithmic focus, application domains, evaluation protocols, and deployment maturity. Second, we diagnose recurring methodological weaknesses, including the use of narrow datasets, superficial evaluation, fragmented metrics, and limited reproducibility. Third, we assess whether the field has achieved sufficient theoretical and methodological progress to support an updated taxonomy, and we propose a new task-based taxonomy. Fourth, we outline a road map for advancing RL in finance, emphasizing reproducibility, task and reporting standardization, realistic data, and transparency. We conclude that while RL in finance is empirically active, it remains conceptually and methodologically immature. Progress requires a shift from fragmented experimentation toward standardized protocols and reproducible research practices.

Keywords: reinforcement learning; financial markets; systematic literature review; methodology

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1. Introduction

In recent years, the application of Reinforcement Learning (RL) to financial markets has emerged as a prominent trend in academic research [1, 2, 3, 4, 5]. This development reflects the growing capabilities of RL algorithms in solving complex decision-making tasks, alongside the demand within financial markets for adaptive, data-driven strategies capable of navigating dynamic and uncertain environments. RL, as a subfield of machine learning, focuses on training agents to make sequential decisions by interacting with an environment and learning from feedback received in the form of rewards [6]. Its theoretical appeal lies in optimizing long-term cumulative outcomes in complex and often stochastic settings. In financial markets, participants—such as traders, portfolio managers, and risk managers—must continuously adapt their strategies based on evolving information. Consequently, RL appears to be a promising method for developing models that can operate in financial domains. At the heart of financial decision-making lies the challenge of making sequential decisions under uncertainty, where each action can influence both immediate outcomes and future opportunities. RL provides a principled approach for addressing such challenges by enabling agents to learn optimal policies through interaction with market environments, either via simulation or historical data analysis. The promise of RL in finance therefore lies in its potential to develop adaptive strategies that can respond effectively to changing market conditions, exploit dynamic patterns, and optimize long-term performance. Nevertheless, realizing this potential demands not only algorithmic innovation but also robust methodological validation—an aspect that has often been overlooked in the existing body of literature.

Previous literature reviews have cataloged a range of RL applications in finance, covering areas such as algorithmic trading, portfolio optimization, market simulation, and risk management [1, 2, 3, 5]. These applications have helped motivate research interest in RL as a tool for solving sequential decision-making problems in finance. This systematic literature review seeks to compliment the existing work focused on cataloging applications by providing a systematic review of the methodological and reporting practices in this field in the period from 2020 to 2024. Our contributions are fourfold: first, we characterize the current state-of-the-art (SOTA) in terms of algorithmic focus, applications, evaluation practices, and deployment maturity; second, we identify recurring methodological weaknesses, including poor reproducibility and superficial evaluation; third, we develop a task-based organizing framework for RL in finance; fourth, based on the derived insights we then introduce a road map to advance the field.

Several motivations drive this work. The number of RL-finance studies has grown rapidly in recent years, signaling an expanding research area. However, persistent methodological limitations have been repeatedly highlighted by prior literature reviews. These include unrealistic market assumptions, lack of robustness to tail risk and extreme events, oversimplified models of market impact and liquidity, reliance on clean and noise-free historical data, short-term optimization bias, lack of standardized benchmarks, challenges related to explainability, and overfitting to specific environments or datasets [2, 3, 5]. Despite these recurring concerns, the methodological standards and empirical reliability of recent literature have not yet been systematically assessed in prior literature reviews. Guided by the aforementioned considerations, we structure our investigation around three central research questions: (1) What is the current state-of-the-art (SOTA) in applying RL to financial problems? (2) How rigorous and reliable are the methodologies employed in recent studies? (3) How can we organize RL in finance research for cumulative progress?

The remainder of this systematic literature review is structured as follows. We begin with introducing RL, its method classes, and its applications in finance in Section 2. Section 3 outlines the data collection and screening process used for the systematic literature review. Section 4 presents our findings. Section 5 discusses the findings and proposes a road map to advance the field. Finally, Section 6 concludes the study and offers recommendations for future research.

2. Preliminaries

This section provides a brief overview of concepts required for the remaining article. We begin by introducing RL methods in Section 2.1. In Section 2.2, we outline the main categories of financial decision-making problems to which RL is typically applied. We conclude with a discussion of the train-test paradigm in different domains in Section2.3.

2.1. Reinforcement Learning: Definitions and Classes

Reinforcement Learning (RL) is a subfield of machine learning concerned with how agents ought to take actions in an environment to maximize cumulative reward [6]. At its core, reinforcement learning formalizes the sequential decision-making problem as a Markov Decision Process (MDP), defined by a tuple $(\mathrm{S},\mathrm{A},P,R,\gamma)$ [6]. Here, S is the set of possible states, A is the set of possible actions, $P(s' \mid s,a)$ denotes the transition probability of reaching state $s' \in \mathrm{S}$ when taking action $a \in \mathrm{A}$ from state $s \in \mathrm{S}$, $R(s,a)$ is the reward received after taking action a in state s, and $\gamma \in [0,1]$ is the discount factor that determines the present value of future rewards. The agent's objective is to learn a policy $\pi(a \mid s)$ that maps states to actions in a way that maximizes the expected cumulative reward over time.

Reinforcement learning methods can be broadly categorized into four classes, each defined by how the agent learns to make decisions, although they are not mutually exclusive [6]. *Value-based methods* aim to learn a value function that estimates the expected cumulative reward of taking a particular action in a given state. The policy is not learned directly but derived implicitly by selecting actions that maximize this estimated value. Classic examples include *tabular Q-Learning* [7] and *Deep Q-Networks (DQN)* [8], which are suitable for discrete action spaces. *Policy-based methods* directly optimize the policy itself. Rather than learning a value function, these methods adjust the parameters of the policy, typically through gradient ascent, to maximize expected returns. A representative example is the classic *REINFORCE* algorithm [9]. *Actor-Critic methods* combine elements of both value-based and policy-based approaches. These methods maintain two components: an *actor*, which learns the policy, and a *critic*, which estimates the value function to guide and stabilize the actor's updates. Notable instances include *A2C* [10], *DDPG* [11], *PPO* [12], and *SAC* [13]. The methods so far are considered *model-free* as they learn directly from the samples and do not make explicit assumptions about the environments transition dynamics. *Model-based methods*, such as MuZero [14] for games, add a further dimension by learning a model of the environment's transition dynamics. This model is used to simulate outcomes or plan action sequences without relying exclusively on real environment interactions. Model-based methods tend to be more sample efficient but require the agent to accurately capture the structure of the environment—an especially demanding task in complex or noisy domains like finance. Each class embodies different assumptions and design choices, offering distinct strengths and limitations in terms of stability, efficiency, and applicability to different types of decision problems.

2.2. Applications of Reinforcement Learning in Finance

The sequential and uncertain nature of financial decision-making—where each action may influence both immediate returns and future opportunities—makes RL seem like a conceptually attractive framework for many financial applications. Hambly et al. (2023) catalog the most common applications for RL in finance [3].[1] *Algorithmic Trading* involves the automatic execution of buy and sell orders based on pre-defined rules or data-driven strategies. The goal is typically to exploit short-term price movements, react to market signals in real time, and optimize execution across fragmented markets. These systems must operate in fast-paced environments where timing and responsiveness are critical. *Portfolio Management* involves allocating capital across a set of assets to achieve investment objectives such as maximizing return, minimizing risk, or maintaining diversification. Portfolio managers must continuously adapt their allocations in response to new market data, economic indicators, or investor preferences, often subject to constraints like risk budgets or transaction costs. *Market Making* focuses on providing liquidity by continuously quoting both buy (bid) and sell (ask) prices for financial instruments. Market makers aim to profit from the bidask spread while managing inventory risk and responding to order

[1] For an in-depth discussion of applications, we refer to [3]

flow. This task requires careful balancing of pricing strategy, risk exposure, and market responsiveness, often under high-frequency conditions. *Optimal Order Execution* deals with the challenge of executing large trades without adversely impacting the market price. Traders must decide how to break up and schedule orders over time to minimize market impact, slippage, and transaction costs. This requires forecasting short-term market movements and adapting to liquidity conditions. *Option Pricing* concerns the valuation of derivative contracts whose payoff depends on the future value of underlying assets. Accurate pricing models must account for uncertainty, volatility, and time to maturity. Related tasks include designing hedging strategies to mitigate risk associated with price movements or volatility shifts.

2.3. Train-Test Paradigm in Finance

The standard empirical machine learning methodology, training a model on historical data and testing it on a holdout set, is widely used across domains such as computer vision, natural language processing, and RL in games and robotics to make claims about the generalizability of machine learning solutions [8, 15]. However, the epistemic value of such an approach depends on the nature of the environment in which learning takes place. In domains like vision, games or robotics, this methodology is appropriate because the structure of the environment is relatively stable, well-understood, and mechanistically modeled. In contrast, applying this methodology in finance introduces a set of domain-specific challenges that could undermine its reliability and interpretability. In image classification, the world being modeled (e.g., handwritten digits, animals, objects) is governed by stable and well-defined features [15]. The test distribution is often assumed to be drawn from the same underlying process as the training distribution [15]. Even when this assumption is violated, these violations can be explicitly studied and understood. Furthermore, errors are typically bounded and interpretable: a cat is likely misclassified as a dog, not as a traffic light. Similarly, in games or robotics, although the environment is dynamic, it is usually engineered, observable, and repeatable [8]. Simulated environments approximate realworld constraints closely, and agents can be tested in real-world settings to close the simulation–reality gap. Failures in performance can be traced back to specific design flaws or incorrect assumptions about physical dynamics, enabling iterative refinement and falsification. In contrast, financial markets cannot be assumed to be stationary, [3, 16, 17]. Patterns that may have existed during a historical training period may no longer hold in the future, particularly if agents begin to exploit them. There is no underlying "ground truth" for structure — only temporarily emergent regularities, often corrupted by noise, regime shifts, and adversarial behavior. Unlike in robotics or vision, we do not have a mechanistic model of the data-generating process, nor do we fully understand which features are causally relevant. In short, while the methodology of train-test evaluation can be applied across domains, its epistemic significance and interpretation differ drastically. In domains with well-defined, stationary environments, this approach allows us to make statements about model generalization. In finance, we cannot assume stationarity for the reasons outlined above. Therefore, even if we apply the same methodology, we cannot interpret empirical results in finance as evidence for learning ability or generalizability in the same way we do in other domains such as vision, games or robotics [18, 19, 20].

3. Methodology and Data for Systematic Literature Review

This section describes how we selected the final corpus of literature and further constructed the dataset used for analysis. We first explain the search strategy and screening protocol used to identify relevant studies. We then provide an overview of the resulting corpus and describe how we extracted structured information for downstream analysis.

To identify relevant literature, we formulated a structured query combining terms related to RL and financial applications (see Appendix A). The query was executed in the Scopus database, focusing on established, state of the art peer-reviewed articles and excluding pre-prints. We further limited the query to English-language journal articles published between January 1st, 2020 and November 29th, 2024, covering the last five years, balancing size of the corpus and recency. The subject areas included Mathematics, Computer Science, Decision Sciences, Economics, Finance, and Business. We applied a two-stage screening process using the Covidence Systematic Review Tool. First, we conducted a title and abstract screening to exclude irrelevant studies. Then, we performed full-text screening to filter studies through our exclusion criteria. We excluded papers that did not explicitly apply RL techniques to financial problems. This process yielded 543 initial hits, of which 163 studies were retained after screening. Figure B.7 in Appendix B presents a PRISMA-style flowchart of the selection process. Appendix C details all studies excluded in full-text screening and their reason for exclusion.

We extracted structured information from each included study using a standardized data extraction template (see Appendix D). The template consists of six sections, each mapped to one of our research questions introduced in Section 1: General Information (RQ1), Dataset Details (RQ2), Generalizability of Results (RQ2), Methodological Rigor (RQ2), Practicality and Benchmarking (RQ2), and Transfer Learning and Taxonomies (RQ3). Each section was broken down into specific questions designed to capture publication characteristics, modeling choices, data properties, evaluation protocols, and transparency practices. For data fields with a categorical format that are not mutually exclusive, we included all categories that apply. This means, for example, that there can be more recorded observations of applications or RL methods than there are number of articles.

Our final corpus consists of 163 studies, including 160 empirical studies and three literature reviews. Figure 1 shows a steady increase in annual publication counts from 2020 to 2024, reflecting growing academic interest in applying RL to finance. The code and dataset used for analysis are available in our GitHub repository[2]. The extracted dataset supports both quantitative and qualitative analysis across methodological, empirical, and conceptual dimensions.

[2] [removed for anonymity – available in author statement]

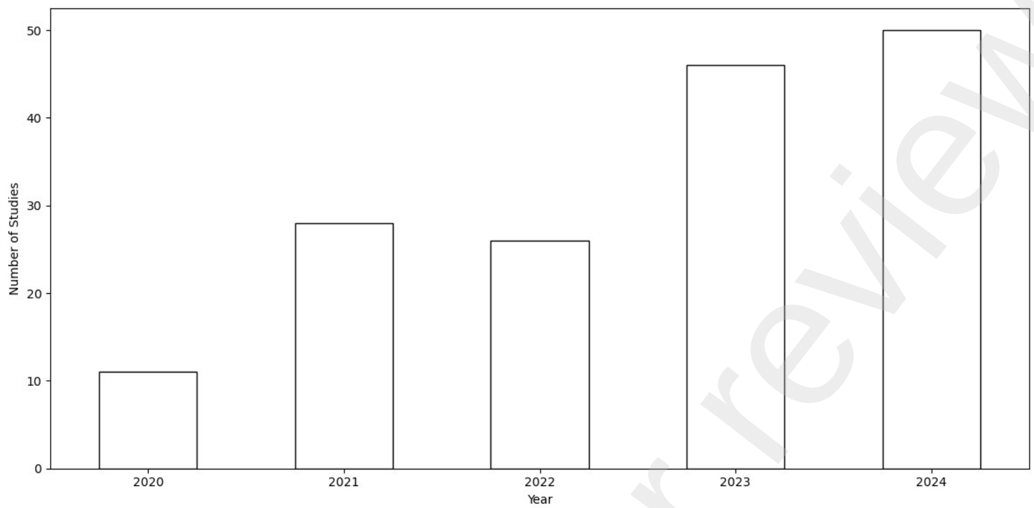

Figure 1: Number of publications in corpus by year. We can see a steady increase in number of publications.

4. Results

We begin with an analysis of co-authorship to visualize collaboration across the corpus in Section 4.1. Section 4.2 discusses applications and RL techniques used. Section 4.3 reviews data diversity, coverage and quality. Section 4.4 highlights benchmark and evaluation practices while Section 4.5 discusses empirical results. We conclude with Section 4.6 analyzing prevailing transparency and reproducibility practices in the corpus.

4.1. Co-Authorship Analysis

To assess the collaborative structure of the research community, we conducted a co-authorship analysis using VOSviewer based on the publication metadata in our review corpus. The resulting network map reveals a highly fragmented landscape, characterized by one central hub and numerous small, disconnected clusters. The central component of the network is anchored around authors such as *Zhang Y.*, *Li J.*, and *Zhang Q.*, who appear frequently across the literature and serve as key nodes within a moderately interconnected group. This cluster represents the only substantial hub of collaboration, where co-authorship links are relatively dense and span multiple papers and authors. In contrast, the majority of authors are distributed across small, isolated clusters, typically composed of two or three researchers, with little to no collaboration outside their immediate group. Examples include clusters centered on authors like *Huang Y.*, *Grilli L.*, or *Park H.*, which remain peripheral to the main network. The absence of cross-cluster collaboration suggests a lack of integration within the field, with researchers often working in separate clusters rather than collaborating. The temporal coloring of the network, where newer publications appear in yellow and older ones in purple, further indicates that recent work has not necessarily contributed to increased integration. Many of the more recent nodes are still appearing in isolation or within disconnected micro-clusters (like *Huang Y.*), suggesting that the fragmentation is persisting rather than diminishing over time.

4.2. Applications and Reinforcement Learning Techniques

Understanding which financial problems researchers focus on, and which RL methods they choose to address them, is crucial for assessing the maturity and direction of the field. The distribution of application domains reveals where RL is perceived as most promising, while the choice of RL techniques reflects assumptions about the underlying environment, available data, and desired learning properties. As we can see in Figure 3, the literature shows a strong focus on Algorithmic Trading and Portfolio Management applications, with 72 studies addressing Algorithmic Trading and 64 studies addressing Portfolio Management. In contrast, Market Making (5 studies), Optimal Order Execution (4 studies), and Option Pricing (2 studies) appear significantly less often. Sixteen studies could not be clearly categorized. In [21], for example, the authors investigated price direction prediction of Bitcoin, while [22] studied the simulation of a cryptocurrency exchange. The focus on Algorithmic Trading and Portfolio Management may be explained by the relative ease of modeling these problems or the presence of more obvious use cases for RL in these domains. As concrete examples, [23] introduces a deep evolutionary RL model for high-frequency trading of digital currencies, demonstrating the application of RL to Algorithmic Trading. In

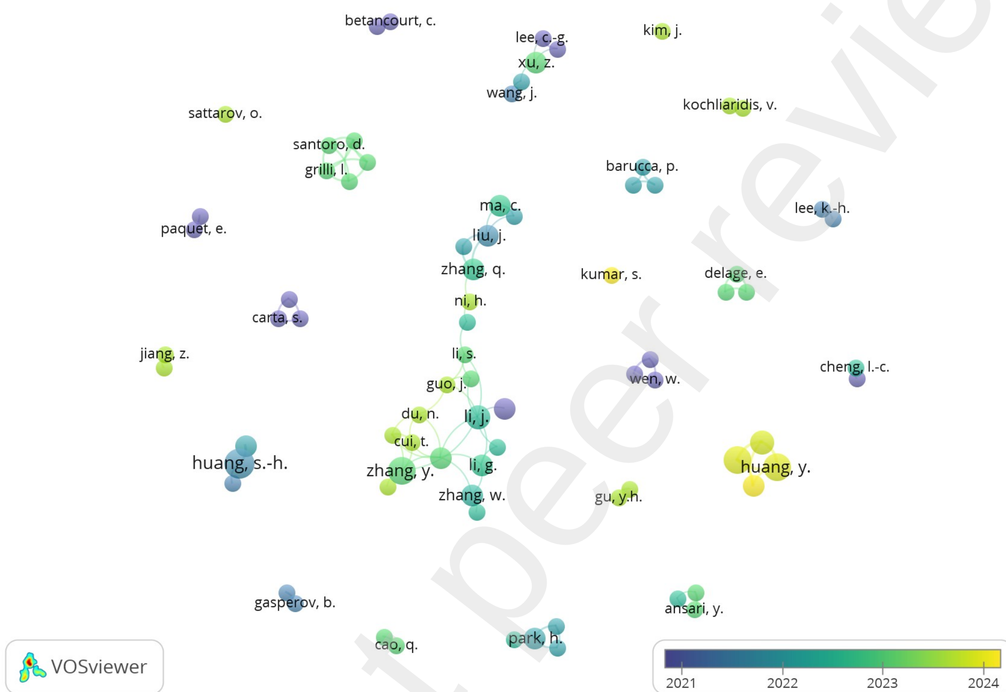

Figure 2: Co-authorship network of RL in finance studies. Node size reflects publication count. Color indicates average publication year (purple = older, yellow = recent). The network reveals one central hub and many small, disconnected clusters.

Portfolio Management, [24] proposes an RL framework integrated with a mean-variance model for portfolio allocation, which dynamically adjusts to an investor's risk preferences.

With respect to RL techniques, the literature is dominated by model-free approaches. Actor-Critic (AC) methods are the most common, appearing in 70 studies, followed by Value-based methods in 64 studies and Policy-based methods in 49 studies. Model-based RL appears in 8 studies, despite its theoretical potential for improving sample efficiency and leveraging structured knowledge about financial markets. The preference for model-free approaches may stem from a perceived difficulty in accurately modeling financial markets or a desire for methods that require fewer structural assumptions. For exam-

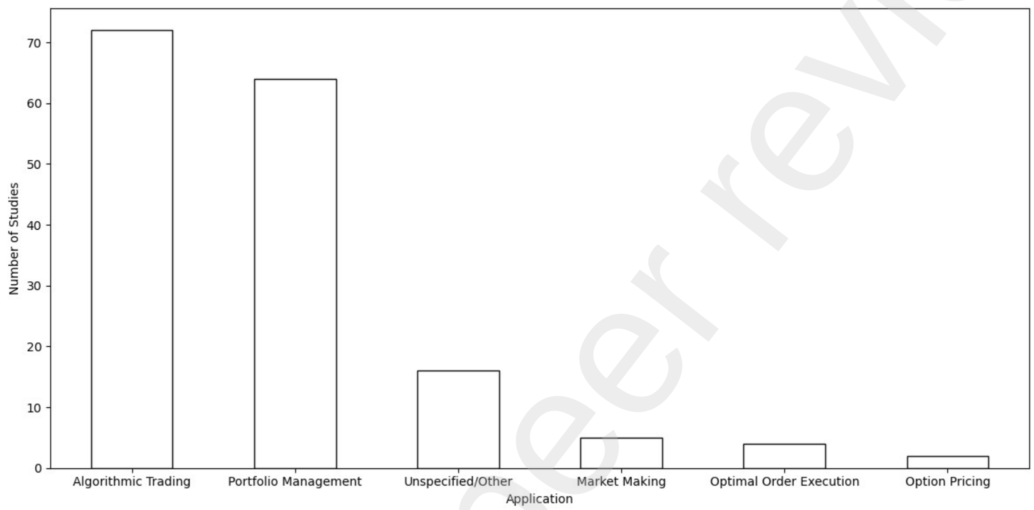

Figure 3: Number of studies by application. We can see that Algorithmic Trading and Portfolio Management are the most common applications. The categories are not mutually exclusive, therefore a study might be counted in multiple categories.

ple, [25] applies a range of AC methods, including A2C, DDPG, and PPO, to high-frequency trading of digital currencies, explicitly comparing the risk preferences of different agents. In the value-based category, [26] implements a discrete event model within a Q-learning framework. As a rare example of model-based RL, [27] specifies transition dynamics for portfolio optimization using non-Gaussian environment models, directly modeling state transitions in a financial context.

The joint distribution of RL techniques and financial applications, shown in Table 1, reveals additional insights into methodological tendencies within the literature. Value-based methods are especially prevalent in Algorithmic Trading

(42 studies), where their simplicity and compatibility with discrete action spaces make them appealing. Actor-Critic methods are also widely used, particularly in Portfolio Management (31 studies) and Algorithmic Trading (30 studies), likely due to their stability in continuous control settings and strong empirical performance. In contrast, model-based RL remains underutilized across all domains, with a few instances observed in Portfolio

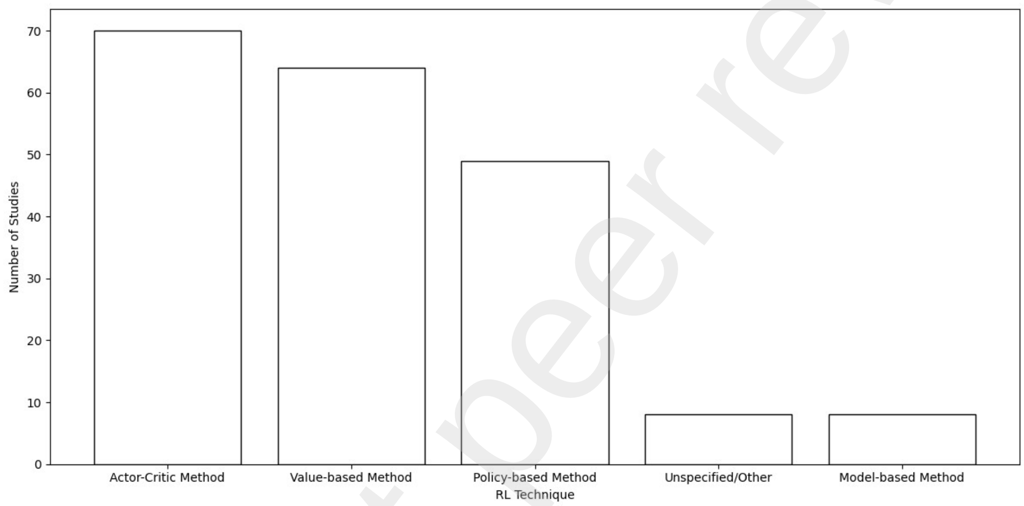

Figure 4: Number of studies by reinforcement learning technique. We can see that modelfree RL methods are most common.

Management (4 studies) and Option Pricing (1 study). Applications such as Market Making and Optimal Order Execution receive minimal attention overall. This pattern reflects a practical bias toward model-free approaches in domains where benchmark procedures and historical datasets might be more readily available. The low representation of model-based methods, despite their theoretical advantages in sample efficiency and planning, may reflect the perceived difficulty of constructing accurate transition models in highdimensional, noisy financial systems. Moreover, the concentration of certain methods in specific applications suggests that methodological exploration remains limited, potentially hindering generalization and the development of more robust RL frameworks for finance.

4.3. Data Diversity, Coverage, and Quality

The literature shows a strong reliance on homogeneous datasets, primarily focusing on equities, developed markets, and daily price data. This narrow scope raises questions about the generalizability of findings beyond standard financial markets. The following evidence supports this observation: 112

Table 1: Number of studies per RL technique and finance application. We can see that most studies are model-free and either Algorithmic Trading or Portfolio Management.

Finance Application	Actor-Critic	Model-based	Other	Policy-based	Value-based
Algorithmic Trading	30	3	5	21	42
Market Making	3	0	0	0	2
Optimal Order Execution	2	0	0	1	3
Option Pricing	1	1	0	0	0
Portfolio Management	31	4	1	25	11
Unspecified/Other	3	0	5	2	6

studies used equity data, followed by 31 on cryptocurrencies, 11 on derivatives, 10 on commodities, and 6 on currencies. Geographically, 118 studies included North American markets, 94 Asia, 60 Europe, and 39 South America or Africa. In terms of data granularity, daily price data dominated with 108 studies, compared to 10 studies using minute data, 14 using order book data, and four using tick-level data. The homogeneity in used datasets might suggest a preference for convenience over representativeness. The focus on homogeneous datasets may limit the ecological validity of findings, restricting their applicability to broader financial environments. Furthermore, the limited use of high-frequency data neglects important real-world market phenomena such as latency, intraday volatility, and market microstructure effects. A typical example supporting this claim is [28], who uses equity indices from the US, Germany, and South Korea with daily data. In contrast, [29] challenges this pattern by using crude oil data across WTI, Brent, and Oman markets, while [23] applies a deep evolutionary RL model for high-frequency digital currency trading, providing counterexamples against the trend.

The reviewed literature also pays limited attention to data quality, often relies on market data alone, and features inconsistent temporal coverage, which may compromise the robustness of empirical findings and heighten the risk of overfitting. Out of 160 studies, 23 explicitly address data quality concerns. Regarding data sources, 62 studies use market data exclusively, while a few incorporate other data types: seven use news, seven use fundamental data, and five use social media. Temporal coverage also varies considerably, with an average of 3810 trading days and a median of 3542, indicating diverse dataset lengths across studies. Neglecting data quality risks undermining result validity. Limited feature sets constrain the RL agents' ability to model complex financial environments, and variable dataset lengths complicate cross-study comparisons. Short samples may inflate performance metrics and increase the risk of overfitting, thereby reducing the reliability of reported outcomes. [30] exemplifies this pattern by relying on one month of tick and volume data without discussing data quality, further testing on 10 trading days. On the contrary, [24] uses 50 Taiwanese index constituents with price and volume data that spans 17 years from 2003 to 2021. To summarize, the limited standardization of data usage and reporting appears to impede cumulative progress across studies.

4.4. Evaluation Practices and Benchmarking

Out-of-sample testing is a widely applied standard across the reviewed literature, with 146 studies implementing some form of it. Among these, 84 studies also tested on a different time series, while 75 studies tested on multiple different timeframes of the same data. A small subset (3 studies) did not split between training and testing data at all. 25 studies evaluated models on entirely new datasets or applications, suggesting that cross-dataset or crossapplication generalization is rarely tested. Furthermore, 33 studies applied statistical significance tests or confidence intervals, raising concerns about the reliability and reproducibility of the reported results. This widespread but superficial adherence to machine learning evaluation standards suggests a field focused on baseline compliance rather than rigorous testing design. The dominant use of narrow evaluation setups limits the robustness of findings and risks undermining confidence in model performance claims. The lack of cross-application testing and the infrequent use of proper statistical validation exacerbate the risk of overfitting and hinder the generalizability of results. Examples from the literature illustrate both good practices and common shortcomings. [31] applied benchmarking and used confidence intervals in evaluation, demonstrating methodological awareness. [32] also benchmarked their models and applied multiple statistical tests across various datasets, including multivariate time series domains like human activity classification and sports activity datasets. In contrast, [30] failed to address data quality concerns, used tick and volume data from one month, and tested on ten trading days. Similarly, [24] applied out-of-sample testing using rolling window splits but omitted statistical validation and confined the study to the Taiwan 50 Index.

Benchmarking is widespread in the literature, with 150 studies including at least one benchmark model. The types of benchmarks vary considerably: 97 studies compared against what they considered state-of-the-art (SOTA) methods, 85 used traditional baselines, 82 included naïve or random benchmarks, and 44 studies used market indices. Furthermore, 149 studies utilized a shared dataset for evaluation purposes. However, a consistent definition of SOTA is often lacking, further complicating the interpretation and comparability of results. The metrics landscape is equally fragmented. A total of 191 distinct metrics were recorded, with the most common being the Sharpe ratio (104 studies), Maximum Drawdown (MDD) (81 studies), and PnL (40 studies). The proliferation of non-standardized metrics significantly impedes the ability to compare results across studies. This not only risks selective reporting but also contributes to a fragmented research landscape, ultimately hindering the accumulation of systematic knowledge in the field. For instance, [33], [34], and [35] use *accumulated return*, [32], [36], [37], and others use *annual return*, [38] reports *annualized percentage return*, and [39, 40] and [41] refer to *annualized rate of return*. Additionally, [42, 43, 44], and others use *annualized return*. Often the studies also lack a definition of the metrics used, further confusing whether the difference between metrics is just in name or also in definition. This variation in terminology illustrates the lack of standardization even for seemingly common metrics.

4.5. Empirical Results

We observe a clear tendency toward transparency of results across the literature, with many studies reporting detailed performance metrics. However, these studies are largely confined to simulation-based environments, with very few studies engaging in real-world validation. Performance claims tend to be modest or mixed, reflecting either cautious interpretation or limited evidential support. Among the reviewed studies, 131 disclosed full performance results, indicating a degree of transparency in reporting. These studies provided numerical outcomes, charts, or detailed evaluation metrics that allowed for a reasonable understanding of the model's behavior across all experimental results. However, 29 studies did not share complete outcomes, potentially omitting critical details relevant for interpreting the reported results. When it comes to real-world deployment, 4 studies provided evidence of testing their models in live trading environments. [30] tested on a demo account for ten days, while [45, 46] reported high-frequency trading deployment for six weeks, claiming significant profit from the strategy and also presented similar results from a live trading test between October and November 2019 in the Chinese commodity market. In contrast, the remaining 155 studies relied entirely on simulations with synthetic or historical data. Finally, 55 studies explicitly claimed that their models outperformed benchmarks, such as indices, buy-and-hold strategies, or approaches that the authors considered as SOTA. The number of studies that claim superior performance is a relatively small proportion of the total corpus, especially considering that most papers proposed new algorithms or adaptations. These statistics indicate that while researchers are generally willing to report results, there remains a substantial gap between current experimental results and desired theoretical performance results as well as practical application. The limited number of strong performance claims might result from inconsistent evaluation standards or lack of statistical validation (see Section 4.4), which weakens the strength of the evidence presented. This disconnect between the ambitions of RL in financial markets and its theoretical advancement as well as actual deployment highlights concerns about the theoretical contributions as well practical relevance of much of the current research.

A second tendency in the literature is the widespread absence of acknowledged limitations of the experimental setups, which might undermine critical reflection on overfitting risks, model validity, and generalizability. Of the 160 studies reviewed, 55 explicitly discussed their methodological or experimental limitations. The absence of such reflections risks overstating the reliability of their findings and hinders the field's scientific maturity. As a positive example, [30] acknowledge several shortcomings, including the use of only one asset, a lack of baseline comparisons, limited testing on a demo account, absence of market impact modeling, and exclusive use of Q-learning without algorithmic comparisons. In contrast, [24] does not include a limitations section. Despite using a deep reinforcement learning model on potentially volatile and historically recent cryptocurrency markets, the study omits discussion of overfitting, robustness over time, transaction costs, market impact, and other theoretical and practical concerns. Moreover, the model is evaluated solely on historical data, yet this limitation is not acknowledged. We emphasize that [24] is not standing out among the studies in the corpus but is an illustrative example for a practice that we observe across the literature. Omitting to discuss limitations adds to a broader pattern in the literature that impedes the assessment of generalizability, reliability and applicability.

4.6. Transparency and Reproducibility

A critical barrier to reproducibility in the reviewed literature is the low level of transparency regarding datasets, source code, and hyperparameter tuning procedures. These components are essential for verifying experimental results, enabling critical peer assessment, and facilitating cumulative scientific progress. However, the analysis reveals that 9 studies directly linked their datasets, 17 made their code available in repositories, and 39 provided a clear explanation of their hyperparameter tuning process. This means that, for most studies, core implementation and experimental details remain inaccessible to other researchers. The implications of this insufficient transparency are significant. Without access to the data or code, it becomes nearly impossible to replicate findings or test their robustness under varied conditions, which might be especially relevant in finance (see Section 2.3). Moreover, when hyperparameter tuning procedures are omitted or not fully explained, it raises concerns about whether potential performance gains are due to changes in model architecture or a more extensive hyperparameter optimization. This lack of methodological clarity undermines the credibility of performance claims and creates obstacles for building upon previous work. For example, [46], despite reporting strong real-world performance, does not share code or data, making independent verification or comparison nearly impossible. In contrast, [30] exemplifies good practice by making both data and code publicly available, thereby supporting reproducibility and transparency.

The vast majority of studies in the reviewed corpus do not meet basic transparency and reproducibility standards as outlined by the *FAIR principles* introduced by [47]. Out of the 17 studies that shared their code, 15 of these could be executed as-is without requiring substantial modification. Even among those, most codebases were narrowly scoped, hardcoded for specific datasets or file paths, and minimally documented, which significantly limits their practical usability. In the other two cases, the provided repositories were either broken or linked to unrelated content, further illustrating the fragility of reproducibility claims in the field. Although the availability of any code is a positive sign, the broader landscape reveals that these codebases are often designed for internal use rather than for reuse by the research community. Limited documentation and rigid implementations make it difficult for other researchers to adapt, extend, or even verify the experiments described in the studies. This suggests an absence of robust community norms around reproducibility and a lack of quality control in the publication process. Notable exceptions include [48] and [49], both of which share not only functional code but also the corresponding datasets, thereby offering fully reproducible pipelines.

Even among the minority of studies that share code, most fall short when it comes to supporting flexibility in data handling and experimental design. Specifically, the majority of submitted codebases do not allow for easy data updates or flexible input integration, both of which are requisites for testing model robustness over time or across different market conditions. Two studies allowed users to re-run experiments with updated market data via APIs, enabling a more realistic evaluation of temporal generalization. An additional 12 studies permitted the use of custom datasets, though in most cases this still required manual file manipulation and came with limited documentation or guidance. This lack of flexibility undermines the ability to evaluate how well a model generalizes beyond its original training window or market regime. Without mechanisms for updating input data or modifying experimental setups, results remain tied to static snapshots. This not only limits the scientific utility of the code but also creates a risk that reported results are overly specific to the conditions under which the experiments were originally run. Two studies stand out as positive examples: [50] and [51] provide a fully executable codebase that allows users to change timeframes and update inputs via an API. These implementations demonstrate how thoughtful design can support both reproducibility and extensibility. We summarize the statistics mentioned in Table 2 and list the full results of the analysis in Table E.13 in Appendix E.

Table 2: Summary of Reproducibility Across 17 Code-Sharing Studies

Reproducibility Criterion	Count	Of 17 Code-Sharing Studies	Of 160 Total Studies
Re-run as is (same code + data)	14	82.4%	8.8%
Re-run with updated price data (same code + updated data)	2	11.8%	1.3%
Re-run with custom data (same code + custom data)	12	70.6%	7.5%
Code available but broken/missing/misleading	2	11.8%	1.3%

5. Discussion

This section discusses the key findings of our review by revisiting the three research questions posed at the outset. We integrate empirical evidence and methodological observations to assess the state of RL in finance and its potential trajectory as a scientific discipline. Based on the diagnosis in Section 5.1 we propose a road map for RL in finance research in Section 5.2.

5.1. Diagnosis

First, we examine the current state of the art and emerging trends. We find that the field is empirically active but methodologically stagnant. The majority of studies focus on applying model-free RL methods to PM and AT in a narrow subset of developed equity markets. Of the 160 studies reviewed, 136 address either PM or AT, and eight use model-based RL (Section 4.2). The dominant algorithms are Actor-Critic (70), Value-based (64), and Policy-based (49). Dataset use is similarly constrained: equities (112 studies), daily data (108 studies), and U.S. markets (118 studies) dominate, with fewer than 10 studies incorporating alternative data sources like news, fundamentals, or social media. We find that 23 studies explicitly address data quality. Evaluation practices are generally weak: 25 studies test generalization on new datasets or applications (Section 4.4), and 33 apply some form of statistical validation such as confidence intervals or significance tests (Section 4.4). Real-world deployment is virtually nonexistent, with 4 studies reporting live implementation (Section 4.5). These findings suggest that while RL in finance research is rich in technical experimentation, it might lack the theoretical depth and methodological maturity needed to build a cumulative body of knowledge.

Turning to the second research question—whether the methodological standards in RL in finance research are rigorous and reliable—we observe widespread inconsistency in the application of evaluation standards. While 146 studies employ some form of out-of-sample testing, 25 extend their evaluation to new datasets or market conditions, and 33 include statistical validation (Section 4.4). We show that 55 studies (34%) report benchmark outperformance, often without robustness checks or significance testing (Section 4.4). Metrics are highly fragmented: 191 unique metrics appear across the corpus, with even widely used ones like cumulative return being inconsistently defined (Section 4.4). Although 150 studies include a benchmark, these are frequently weak. On data quality and generalizability, most studies rely on homogeneous equity data from developed markets, and 23 discuss data quality explicitly. Very few attempt to test generalization across different time periods, applications, or market types. Transparency is similarly limited: 17 out of 160 studies provide code (Section 4.6). Reproducibility is compromised by rigid, undocumented codebases while 2 studies allow rerunning experiments with updated market data via an API (Section 4.6). Methodological advancement might be necessary for the field to progress toward cumulative and generalizable knowledge.

Finally, we assess whether the field has developed any organizing frameworks or taxonomies to structure its growth. Existing reviews, such as those by [3], [2], and [5], all classify studies by applications such as Portfolio Management and Algorithmic Trading. While some make attempts to define application types or evaluation criteria, these are loosely connected and derived from industry practices. Building on the task definitions by [20], we propose a taxonomy that organizes the RL literature for finance into three fundamental decision problems—*Prediction*, *Allocation*, and *Execution*—and maps the traditional finance application areas to these tasks. The aim is to expose the underlying decision structure that unifies seemingly diverse financial use cases and to provide a common framework for their comparison. Financial applications such as portfolio management, algorithmic trading, market making, option pricing, optimal order execution, and smart order routing are usually presented as distinct subfields, yet each ultimately reduces to one—or a combination—of the three tasks. Prediction produces forecasts of returns, volatility, or price direction but does not itself generate profit and loss. Allocation determines portfolio weights or positions, directly controlling exposure and risk. Execution converts desired positions into market orders while minimizing cost, market impact, and slippage. Separating applications into these task categories clarifies both methodological requirements and evaluation metrics.

Table 3 summarizes how the finance applications in our corpus map to the three RL task types. Algorithmic trading, portfolio management, and market making are primarily allocation problems in which the agent determines dynamic positions or portfolio weights to maximize risk-adjusted returns or to manage inventory. Price-direction forecasting represents a pure prediction task in which the model outputs probabilistic estimates of future price movement or volatility without direct trading. Option-pricing and hedging studies generally fall under allocation because the RL agent chooses hedge ratios or replicating portfolios to track option payoffs. Optimal order execution is an execution problem in which the agent schedules trades to minimize implementation shortfall for a fixed target quantity. Smart order routing is likewise an execution problem, but the agent must also allocate orders across multiple venues over time to reduce total execution cost. We provide further formalization of the tasks in Appendix G. Applying this taxonomy to the full corpus of studies provides several benefits. It allows us to quantify research concentration by showing, for instance, that 143 studies address allocation while far fewer tackle prediction (13) or execution (4), as shown in Figure 5. It reveals that the prediction and execution tasks remain comparatively underexplored. The taxonomy also bridges disciplinary language: finance practitioners can locate their familiar applications within the taxonomy, while RL researchers understand the underlying decision problems.

Table 3: Mapping of finance application areas to RL task categories.

Finance Application	Task	Typical RL Output
Directional or Numerical Prediction of a Variable	Prediction	Probabilistic forecasts of future price movement or volatility.
Algorithmic Trading (AT)	Allocation	Dynamic long/short or position-sizing decisions aimed at maximizing cumulative return.
Portfolio Management (PM)	Allocation	Multi-asset portfolio weights balancing return, risk, and transaction costs.
Market Making (MM)	Allocation	Inventory and quoting policies that allocate capital between inventory risk and spread capture.

Option Pricing / Hedging	Allocation	Continuous hedge-ratio or replicating-portfolio decisions to track option payoffs.
Optimal Order Execution (OEE)	Execution	Time-sliced order sizes that minimize implementation shortfall for a fixed target quantity.
Smart Order Routing (SOR)	Execution	Venue-specific order allocations over time to minimize total execution cost.

5.2. A Road Map for RL in Finance

Based on our diagnosis, we propose a road map for RL in finance that addresses the field's methodological shortcomings and sets out a pathway

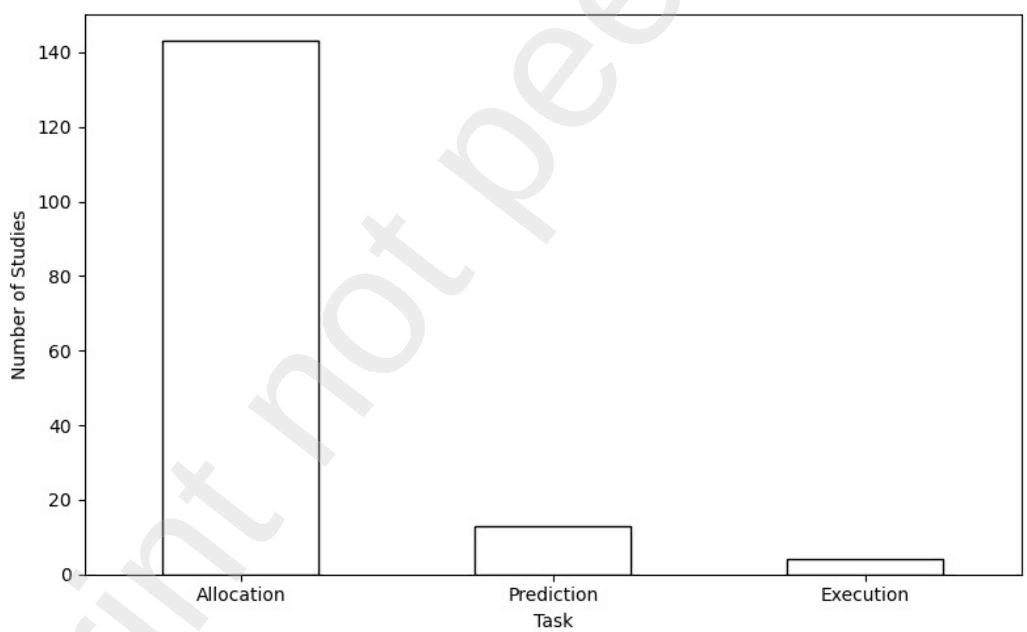

Figure 5: Number of studies by task. We can see that the allocation task is most studied.

toward cumulative scientific progress. The road map links each step explicitly to the evidence gathered in our review, prioritizes actions by their urgency and readiness, and situates them within both the broader RL literature and established practices in quantitative finance. By structuring the challenges into reproducibility, task standardization, evaluation, data, realism, and taxonomy, the road map provides actionable recommendations, illustrates them with parallels from other domains (e.g., Atari benchmarks in RL, reproducibility checklists in ML conferences, or deployment pipelines in algorithmic trading), and highlights how finance-specific contributions can complement general ML standards. We summarize the road map in Table 4.

Table 4: Road map for Advancing RL in Finance: Problems, Solutions, and Examples from Other Fields.

Step	Topic	Problems & Solutions	Examples from Other Fields
1	Reproducibility	Problems: Few codebases shared; poor documentation; few datasets shared. Solutions: Open-source code and data; clear environment setup; commented code; project-level README; "one-run" reproducibility goal.	Reproducibility Checklists (NeurIPS, ICLR); RL Unplugged; Stable-Baselines3 implementations; Hugging Face model hub
2	Task Standardization	Problems: Ad-hoc, applicationspecific task definitions; unclear input–output and reward structures; no shared baselines. Solutions: Define a standardized suite of tasks (e.g., prediction, allocation, execution) with clear specifications and baseline implementations.	Atari and MuJoCo benchmarks (RL); GLUE/SuperGLUE (NLP)
3	Reporting Standardization	Problems: Varying reporting formats; lacking statistical validation; fragmented metrics; hard to compare results. Solutions: Standardize reporting; establish shared reporting protocols.	Datasheet for datasets; Model Cards
4	Data	Problems: Reliance on NA and EU equities; limited asset classes; little use of high-frequency or alternative data; validating data quality is rare. Solutions: Develop libraries for synthetic data generators and data quality validation.	Great Expectations data quality library

5	Realism	Problems: Reliance on simulations; almost no live or paper trading evidence. Solutions: Define a staged evaluation process (simulation → paper trading → shadow testing → deployment); encourage industry collaboration.	Robotics sim-to-realdevelopment pipelines
6	Taxonomy	Problems: Research remains fragmented and hard to synthesize across applications and methods. Solutions: Develop a taxonomy that groups tasks, solution strategies, and evaluation practices into a unified framework.	Task and method taxonomies in ML and robotics

The first step toward strengthening RL research in finance is improving transparency and reproducibility. Our review shows that only a small fraction of studies share code (17 of 160), and even fewer provide datasets or adequate documentation. Where code is available, it is often incomplete, hardcoded for specific data, or insufficiently documented, making it difficult for other researchers to replicate or extend results. This lack of reproducibility undermines trust in reported findings and prevents the accumulation of scientific knowledge. Without open and verifiable research pipelines, each study remains an isolated contribution rather than part of a broader body of evidence. To address this, we propose an *RL in Finance Reproducibility Checklist*, inspired from NeurIPS reproducibility checklists and aligned with the FAIR (Findable, Accessible, Interoperable, Reusable) principles [47]. The full checklist is provided in Appendix F. Concretely, future studies should be:

- Findable: Assign persistent identifiers (e.g., Zenodo DOIs for GitHub releases), provide rich metadata (e.g., data coverage, adjustments, preprocessing scripts), and explicitly link dataset versions.
- Accessible: Ensure code and data are retrievable via open protocols (e.g., GitHub, Hugging Face hub), and if data licenses prohibit sharing, provide preprocessing scripts and synthetic substitutes.

- Interoperable: Use standard formats (CSV/Parquet for data, YAML/ JSON for configs), adopt community vocabularies, and ensure crossreferencing between configs, datasets, and code.

- Reusable: Provide documentation (README, comments, run instructions), clear licensing (e.g., MIT/Apache 2.0 for code, CC-BY 4.0 for synthetic datasets), provenance information (e.g., corporate action adjustments, cleaning scripts), and address finance-specific biases (survivorship, look-ahead, transaction costs). A useful target is *"onerun reproducibility,"* where all results can be reproduced by executing a single script or notebook.

These measures are technically feasible and already standard in other fields. For instance, NeurIPS and ICLR require reproducibility checklists, while initiatives such as *RL Unplugged*[3], *Stable-Baselines3*[4], and the *Hugging Face model hub*[5] show how open datasets, standardized implementations, and unified repositories can foster cumulative progress. In finance, comparable initiatives could include open repositories of financial data and enforced reproducibility checklists in journals and conferences. Establishing these norms would not only enhance the credibility of individual studies but also lay the foundation for cumulative and generalizable knowledge for RL in finance.

The second step requires to standardize tasks. Other fields have demonstrated how task standardization can guide research efforts. *OpenAI Gym* [52] established a structured suite of tasks (e.g., navigation, control, planning), and in NLP the GLUE and SuperGLUE benchmarks [53, 54] group problems by decision type, enabling systematic comparisons and guiding methodological development. For RL in finance, we propose a standardization of tasks following the suggestion of [20] and our proposed taxonomy in Section 5.1. Establishing these tasks does not only organize the field but also anchor research in more

[3] https://github.com/huihanl/rl$_u$*nplugged*
[4] https://github.com/DLR-RM/stable-baselines3
[5] https://huggingface.co/models

standardized problem definitions. As the methodological maturity of the field improves, a taxonomy based on this task distinction could help consolidate cumulative knowledge and guide future exploration. Further specification of the tasks can be found in Appendix G.

The third step for advancing RL in finance is the standardization of reporting practices. Our review reveals that current studies are hindered by fragmented and inconsistent reporting across three dimensions: data, models, and evaluation. Datasets are often insufficiently described, making it difficult to assess their quality, coverage, or potential biases (see Section 4.3). Models are frequently underspecified, with implementation details not shared (see Section 4.6). Evaluation practices are especially heterogeneous: we identified 191 distinctly named performance metrics in the literature, many of which are inconsistently defined, rarely subjected to statistical validation, and often applied to single assets or time periods (see Section 4.4). This lack of transparency undermines reproducibility, encourages selective reporting, and makes it challenging to compare results across studies. Other areas of machine learning have addressed these challenges by introducing structured reporting standards. For instance, *Model Cards* [55] provide standardized templates for documenting model design, assumptions, and limitations, while *Datasheets for Datasets* [56] establish norms for describing dataset characteristics, coverage, and quality. In natural language processing, initiatives such as GLUE and SuperGLUE have complemented such documentation practices with standardized evaluation protocols, enabling transparent comparisons and driving rapid progress. To achieve comparable progress in RL for finance, we argue that the field must adopt standardized reporting. We propose a trifecta of reporting artifacts: a *Data Card*, a *Model Card*, and a *Evaluation Card*. Together, these cards would create a transparent, comparable, and reproducible standard report for what data was used, what model was built, and how it was evaluated, thereby reducing fragmentation and increasing confidence in the reported results. The cards should be seen as minimum reporting standard that can be extended where required. We provide templates for each of the three cards in Appendix H, Appendix I, and Appendix J.

The fourth step is addressing the narrow focus on homogeneous datasets. In Section 4.3, we show that the majority of studies rely on equities from North American and European markets, often using daily price data. Few studies incorporate emerging markets, high-frequency data, or alternative data sources such as fundamentals, sentiment, or order book information. This might limit ecological validity and raise concerns about overfitting to convenient but unrepresentative datasets. In broader machine learning, progress has often been driven by the availability of diverse and open datasets. OpenML repositories, Kaggle competitions, the Hugging Face dataset hub, and classic UCI datasets have played central roles in enabling broader generalization and replicability. Such repositories not only provide diversity but also lower the barrier to entry for researchers and facilitate cumulative evaluation across shared datasets. For RL in finance, comparable initiatives would involve the creation of open repositories of financial time series spanning multiple asset classes (equities, commodities, FX, crypto), regions (developed and emerging markets), and data granularities (tick, intraday, daily). These repositories could be accompanied by *metadata cards* that document data coverage, preprocessing, and known biases, in line with FAIR data management principles. Vendor datasets such as CRSP/Compustat, TAQ, LOBSTER, CME futures, CLS FX data, or alternative sources like RavenPack (news) and Refinitiv (fundamentals) could serve as starting points, complemented by open-access sources such as Binance (crypto) or BIS FX statistics. To realize this vision, we propose time series generators, that provide standardized simulation benchmarks where the mechanism of the data-generating process is fully known. These generators allow controlled experiments to test whether RL agents can distinguish genuine patterns from noise, and they provide baseline environments independent of messy financial data. While synthetic price generators (e.g., agent-based market models, geometric brownian motion, jump diffusion and others) have long been used for stress-testing and ad-hoc RL experiments, these efforts are typically customized, one-time, and may lack standardization [57, 58, 59, 60, 61, 62, 63, 64].

As a first step to operationalize this, we created *[removed for anonymity]*[6] ([removed for anonymity]). Such ground truth generators are analogous to synthetic datasets in ML (e.g., OpenML generators, RL Unplugged) and support the first stage of a deployment ladder, which we will introduce below.

The fifth step addresses the gap between simulation and real-world deployment. Our review finds that nearly all studies rely exclusively on simulated backtests, with four providing evidence of live trading or paper-trading experiments. This overreliance on simulations risks inflating performance claims due to results that might not persist in real-world implementations. We propose a *deployment ladder*, mirroring best practice in quantitative finance and RL (see Figure 6. In this framework, models are first tested on *ground-truth generators* to validate basic learning ability, then on *historical datasets* using walk-forward or rolling validation. Promising candidates progress to *paper-trading platforms* (e.g., Interactive Brokers, QuantConnect, Alpaca), where orders are executed in real market conditions but without financial risk. Next comes *shadow testing*, where RL agents run in parallel with live systems and generate a "shadow PnL" without executing trades. Only after passing these steps should models be considered for *limited-stakes live deployment*, and ultimately *full-scale deployment* with monitoring and risk controls. This staged process is based on already established best practices in both academic finance literature and industry workflows. [65] outline the "K|V" process, which follows the staged progression from idea to prototype to backtest to paper trading and then production. Similarly, [18, 19] emphasize the risks of backtest overfitting and highlight the need for multiple layers of out-of-sample and live validation. In [20], the author further advocates for rigorous walk-forward validation, paper trading, and live testing before capital is deployed. The staged evaluation model is also used in industry. Broker platforms such as Interactive Brokers, QuantConnect, and Alpaca provide dedicated *paper-trading APIs* that allow agents to interact with real-time markets without financial risk, explicitly formalizing the simulation

[6] [removed for anonymity – available in author statement]

to paper trading to live deployment pipeline. Moreover, quant practitioner guides (e.g., QuantInsti training materials [7], Medium quant trading series[8]) frequently describe workflows that include simulation, paper trading, shadow testing, and phased live deployment, mirroring our proposed ladder structure. By adopting a similar maturity model, RL in finance research could report progress in terms of deployment maturity rather than isolated backtest performance, providing a clearer pathway toward practical impact. Progress towards practical application could also surface new research questions to be addressed in earlier stages, therefore academia could guide practice and vice versa. Even without industry collaboration for full deployment, the widespread availability of paper-trading APIs offers an accessible intermediate step to move beyond toy simulations for a more realistic evaluation at the third step of the deployment ladder. We visualized the deployment ladder in Figure 6.

[7] https://blog.quantinsti.com/

[8] https://medium.com/coinmonks/a-beginners-guide-to-quantitative-trading-quanttrading-how-to-get-started-and-succeed-c35fc67c6bb5

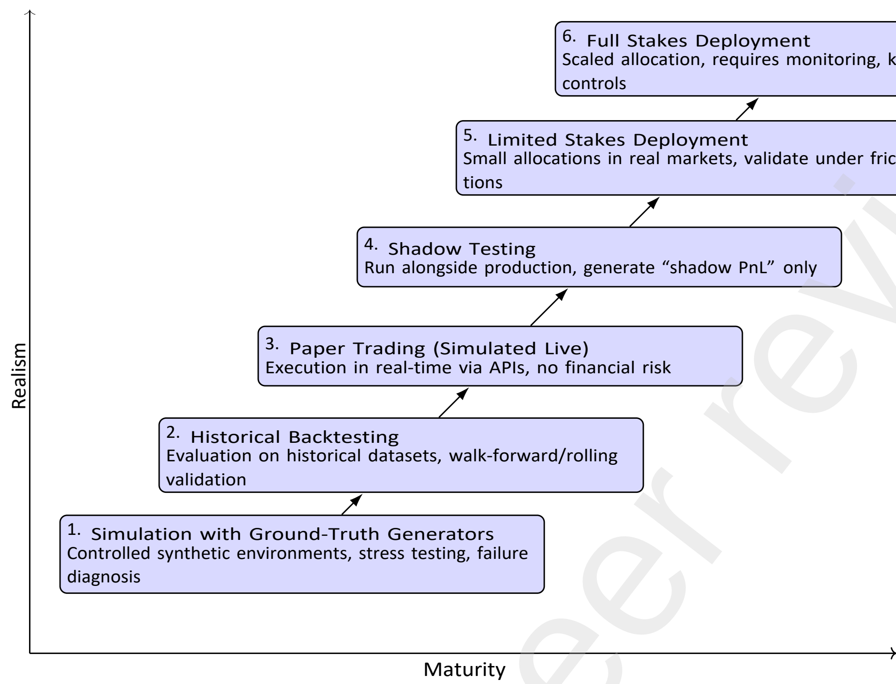

Figure 6: Deployment Ladder for RL in Finance. The framework increases both maturity (x-axis) and realism (y-axis) as models progress from simulation to live trading.

Finally, once reproducibility, standardized tasks, consistent reporting, diverse data, and a staged evaluation pipeline are in place, the field can support a taxonomy that moves beyond application lists. We propose a task-based taxonomy built on three fundamental decision problems outlined by [20]—*Prediction*, *Allocation*, and *Execution*—and suggest to enrich it by *solution strategies* and *evaluation protocols* observed in the literature. Each study could then be cross-classified by its *method family*, and the *evaluation practice* (ground-truth generator, historical backtest, paper trading, live deployment). Such a taxonomy—*tasks* × *solutions* × *evaluation*—parallels mature frameworks in robotics and general RL (e.g., control/navigation/planning) and provides a consistent scaffold for synthesizing results, identifying underexplored intersections, and guiding cumulative scientific progress.

6. Conclusion

This review assessed the state of reinforcement-learning research in finance from the period 2020 to 2024 by analyzing 163 peer-reviewed studies. First, we mapped the field's current practice, finding that research is dominated by model-free algorithms (Actor–Critic, Value- and Policy-based methods) applied primarily to equity markets for the allocation task and evaluated almost exclusively in simulated back-tests, with little evidence of live deployment. Second, we identified recurring methodological weaknesses, including limited data and code sharing, inconsistent or absent reporting of hyper-parameters, and a heavy reliance on the train–test paradigm despite the potentially non-stationary nature of financial markets, which undermines claims of generalizability. Third, we proposed a principled taxonomy grounded in decision structure, grouping applications into the fundamental RL tasks of prediction, allocation, and execution, thereby moving beyond adhoc, application-specific labels. Fourth, we set out a detailed road map for advancing the field, emphasizing (i) rigorous reproducibility through FAIR-aligned code and data release, (ii) task standardization using the proposed taxonomy, (iii) unified reporting standards for data, models, and evaluation metrics, (iv) broader and better-documented data coverage, (v) greater realism via a staged "deployment ladder" from synthetic generators to live trading, and (vi) creating a taxonomy of RL research in finance.

Several limitations of this review must be acknowledged. We did not perform a formal quality appraisal of individual studies, and our inclusion criteria were restricted to peer-reviewed work indexed in Scopus, which may exclude relevant preprints or grey literature. Data extraction necessarily involved subjective interpretation, and the heterogeneity of evaluation protocols precluded meta-analysis. Moreover, the lack of standardized reporting across much of the corpus limited direct comparability of results.

Despite its current shortcomings, RL in finance might be able to advance if the field embraces a more principled scientific approach as outlined in our road map. We therefore advocate for a shift for RL in finance from fragmented experimentation toward cumulative science. By strengthening reproducibility, adopting methodological standards, and building benchmarks, the field may move closer to realizing the potential of RL as a framework for adaptive financial decision-making.

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2023.122801.

Appendix A. Search Query

(TITLE-ABS-KEY "Reinforcement Learning*" OR rl* OR "Deep Reinforcement Learning*" OR "Actor-Critic*" OR "Policy Gradient*" OR "Q-Learning*")

AND

(TITLE-ABS-KEY finance* OR "Financial Market*" OR investment* OR trading*
OR "Portfolio Management*" OR "Risk Management*" OR pricing*
OR "Asset Allocation*" OR cryptocurrency* OR "Algorithmic Trading*")

AND

(TITLE-ABS-KEY "State of the Art" OR sota OR trend* OR benchmark*

```
   OR "Current Practice*" OR development* OR framework*
   OR classification* OR taxonomy* OR organization* OR "Categorization*"
   OR "Research Map*" OR "Systematic Review*")
AND
(PUBYEAR > 2019)
AND
(LIMIT-TO(SUBJAREA,"MATH") OR LIMIT-TO(SUBJAREA,"DECI")
   OR LIMIT-TO(SUBJAREA,"COMP") OR LIMIT-TO(SUBJAREA,"ECON")
   OR LIMIT-TO(SUBJAREA,"BUSI"))
AND
(LIMIT-TO(DOCTYPE,"ar"))
AND
(LIMIT-TO(LANGUAGE,"English"))
```

Appendix B. Prisma-style Flow Chart of Selection Process

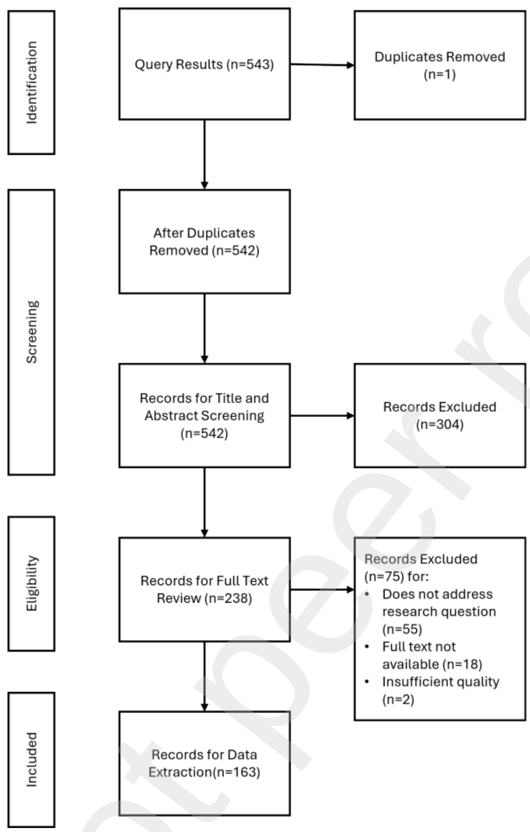

Figure B.7: PRISMA-style flow diagram visualizing the selection process. We included 163 records in the final corpus after the screening process.

Appendix C. List of Studies Excluded in Full-text Review

Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
1	Application Of Deep Reinforcement Learning In Asset Liability Management	Wekwete, T.A.; Kufakunesu, R.; van Zyl, G.	2023	Does not address research questions
2	Short-Term Electricity Futures Investment Strategies For Power Producers Based On Multi-Agent Deep Reinforcement Learning	Wang, Y.; Shi, E.; Xu, Y.; Hu, J.; Feng, C.	2024	Does not address research questions
3	A Novel Stock Portfolio Model Based On Deep Reinforcement Learning	Li, H.; Hai, M.; Zhang, Y.; Li, P.	2021	Full-text not available
4	Optimal Coordination For Multiple NetworkConstrained Vpps Via Multi-Agent Deep Reinforcement Learning	Liu, X.; Li, S.; Zhu, J.	2023	Does not address research questions
5	Deep Reinforcement Learning For Portfolio Management	Ma, Y.; Liu, Z.; McAllister, C.D.	2023	Full-text not available
6	Reinforcement Learning Techniques For Stock Trading: A Survey Of Current Research	Borkar, S.; Jadhav, A.	2024	Full-text not available
7	Deep Q-Learning For Trading Cryptocur-rency	Ma, Y.C.; Wang, Z.; Fleiss, A.	2021	Full-text not available

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Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
8	Portfolio Optimization In Dynamic Markets: Reinforcement Learning For Investment	Sarkar, P.; Dwibedi, P.; Deore, S.S.; Gawali, T.; Kulkarni, M.D.; Saha, A.	2024	Full-text not available
9	Deep Reinforcement Learning For Option Replication And Hedging	Du, J.; Jin, M.; Kolm, P.N.; Rritter, G.; Wang, Y.; Zhang, B.	2020	Full-text not available
10	Islamic Stock Portfolio Optimization Using Deep Reinforcement Learning	Faturohman, T.; Nugraha, T.	2022	Full-text not available
11	Deep Reinforcement Learning In Cryptocurrency Trading: A Profitable Approach	Tay, X.H.; Lim, S.M.	2024	Full-text not available
12	Deep Reinforcement Learning-Based Trading Strategy For Load Aggregators On Price-Responsive Demand	Yang, G.; Du, S.; Duan, Q.; Su, J.	2022	Does not address research questions
13	Flexible Recommendation For Optimizing The Debt Collection Process Based On Customer Risk Using Deep Reinforcement Learning	Sivamayilvelan, K.; Rajasekar, E.; Vairavasundaram, S.; Balachandran, S.; Suresh, V.	2024	Does not address research questions

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Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
14	Optimization Of PeerTo-Peer Energy Trading With A Model-Based Deep Reinforcement Learning In A NonSharing Information Scenario	Uthayansuthi, N.; Vateekul, P.	2024	Does not address research questions
15	Applying Artificial Intelligence In Cryptocurrency Markets: A Survey	Amirzadeh, R.; Nazari, A.; Thiruvady, D.	2022	Does not address research questions
16	Blockchain-Enabled Resource Trading And Deep Reinforcement Learning-Based Autonomous Ran Slicing In 5G	Boateng, G.O.; Ayepah-Mensah, D.; Doe, D.M.; Mohammed, A.; Sun, G.; Liu, G.	2022	Does not address research questions
17	Trend Following Deep Q-Learning Strategy For Stock Trading	Chakole, J.; Kurhekar, M.	2020	Full-text not available
18	Deep Reinforcement Learning Based On Balanced Stratified Prioritized Experience Replay For Customer Credit Scoring In Peer-To-Peer Lending	Wang, Y.; Jia, Y.; Fan, S.; Xiao, J.	2024	Does not address research questions

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Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
19	Portfolio Management Framework For Autonomous Stock Selection And Allocation	Kim, J.-S.; Kim, S.-H.; Lee, K.-H.	2022	Does not address research questions
20	Optimizing Credit Limit Adjustments Under Adversarial Goals Using Reinforcement Learning	Alfonso-Sánchez, S.; Solano, J.; Correa-Bahnsen, A.; Sendova, K.P.; Bravo, C.	2024	Does not address research questions
21	A2Gan: A Deep Reinforcement-Based Learning Algorithm For Risk-Aware In Finance	Le, T.P.; Rho, C.; Min, Y.; Lee, S.; Choi, D.	2021	Does not address research questions
22	Stock Market Prediction Based On Big Data Using Deep Reinforcement Long Short-Term Memory Model	Ishwarappa, K.; Anuradha, J.	2022	Insufficient Quality
23	Deep Adaptive GroupBased Input Normalization For Financial Trading	Nalmpantis, A.; Passalis, N.; Tsantekidis, A.; Tefas, A.	2021	Does not address research questions
24	Deep Hedging Of LongTerm Financial Deriva- tives	Carbonneau, A.	2021	Does not address re- search questions

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No.	Title	Authors	Year	Reason for Exclusion

Table C.6: List of Studies Excluded in Full-text Review and Reasons

25	A New Hybrid Method Of Recurrent Reinforcement Learning And Bilstm For Algorithmic Trading	Huang, Y.; Song, Y.	2023	Does not address research questions
26	An Automated Deep Reinforcement Learning Pipeline For Dynamic Pricing	Afshar, R.R.; Rhuggenaath, J.; Zhang, Y.; Kaymak, U.	2023	Does not address research questions
27	A Reinforcement Learning Approach For The Continuous Electricity Market Of Germany: Trading From The Perspective Of A Wind Park Operator	Lehna, M.; Hoppmann, B.; Scholz, C.; Heinrich, R.	2022	Does not address research questions
28	A Deep Reinforcement Learning Framework For Continuous Intraday Market Bidding	Boukas, I.; Ernst, D.; Théate, T.; Bolland, A.; Huynen, A.; Buchwald, M.; Wynants, C.; Cornélusse, B.	2021	Does not address research questions
29	Developing A Risk Management Approach Based On Reinforcement Training In The Formation Of An Investment Portfolio	Martovytskyi, V.; Argunov, V.; Ruban, I.; Romanenkov, Y.	2023	Does not address research questions

Continued on next page

Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
30	Scalable Reinforcement Learning-Based Neural Architecture Search	Cassimon, A.; Mercelis, S.; Mets, K.	2024	Does not address research questions
31	Risk Budgeting Portfolio Optimization With Deep Reinforcement Learning	Han, S.	2023	Full-text not available
32	Enhancing Stock Trading Strategies: Integrating Discrete Wavelet Transformation With Deep Q-Network	Wang, Q.; Zhang, L.; Zeng, Y.; Wu, S.; Yu, C.; Sun, S.	2024	Full-text not available
33	Profit-Based Deep Architecture With Integration Of Reinforced Data Selector To Enhance Trend-Following Strategy	Li, Y.; Zheng, Z.; Dai, H.-N.; Wong, R.C.-W.; Xie, H.	2023	Does not address research questions
34	Free Market Of MultiLeader Multi-Follower Mobile Crowdsensing: An Incentive Mechanism Design By Deep Reinforcement Learning	Zhan, Y.; Liu, C.H.; Zhao, Y.; Zhang, J.; Tang, J.	2020	Does not address research questions
35	A Learning-Based Incentive Mechanism For Federated Learning	Zhan, Y.; Li, P.; Qu, Z.; Zeng, D.; Guo, S.	2020	Does not address research questions

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Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
36	An Enhanced Wasserstein Generative Adversarial Network With Gramian Angular Fields For Efficient Stock Market Prediction During Market Crash Periods	Ghasemieh, A.; Kashef, R.	2023	Does not address research questions
37	Can Artificial Intelligence (Ai) Manage Behavioural Biases Among Financial Planners?	Hasan, Z.; Vaz, D.; Athota, V.S.; Désiré, S.S.M.; Pereira, V.	2023	Does not address research questions
38	A Search-Based Testing Approach For Deep Reinforcement Learning Agents	Zolfagharian, A.; Abdellatif, M.; Briand, L.C.; Bagherzadeh, M.; Ramesh, S.	2023	Does not address research questions
39	Importance Of Machine Learning In Making Investment Decision In Stock Market	Prasad, A.; Seetharaman, A.	2021	Does not address research questions
40	Qlbs: Q-Learnerinthe Black-Scholes(-Merton)Worlds	Halperin, I.	2020	Full-text not available

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No.	Title	Authors	Year	Reason for Exclusion

Table C.6: List of Studies Excluded in Full-text Review and Reasons

41	A Scalable PrivacyPreserving Multi-Agent Deep Reinforcement Learning Approach For Large-Scale PeerTo-Peer Transactive Energy Trading	Ye, Y.; Tang, Y.; Wang, H.; Zhang, X.-P.; Strbac, G.	2021	Does not address research questions
42	A Novel Linear-ModelBased Methodology For Predicting The Directional Movement Of The Euro-Dollar Exchange Rate	Argotty-Erazo, M.; BlázquezZaballos, A.; Argoty-Eraso, C.A.; LorenteLeyva, L.L.; Sánchez-Pozo, N.N.; PeluffoOrdóñez, D.H.	2023	Does not address research questions
43	Forecasting Stochastic Time Series Using Reinforcement Learning	Boyko, N.; Kachmaryk, V.	2024	Full-text not available
44	Deep Reinforcement Learning For Trading	Zhang, Z.; Zohren, S.; Roberts, S.	2020	Full-text not available

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No.	Title	Authors	Year	Reason for Exclusion

Table C.6: List of Studies Excluded in Full-text Review and Reasons

45	Self-Learning Agents For Recommerce Markets	Groeneveld, J.; Herrmann, J.; Mollenhauer, N.; Dreeßen, L.; Bessin, N.; Tast, J.S.; Kastius, A.; Huegle, J.; Schlosser, R.	2024	Does not address research questions
46	Automated Financial Time Series Anomaly Detection Via Curiosity-Guided Exploration And SelfImitation Learning	Cao, F.; Guo, X.	2024	Does not address research questions
47	A New Reinforcement Learning Approach For Improving Energy Trading Management For Smart Microgrids In The Internet Of Things	Lu, Q.; Li, H.; Zheng, J.; Qin, J.; Yang, Y.; Li, L.; Jiang, K.	2023	Does not address research questions
48	Application Of Two-Dimensional Fibonacci Wavelets In Fractional Partial Differential Equations Arising In The Financial Market	Sabermahani, S.; Ordokhani, Y.; Rahimkhani, P.	2022	Does not address research questions

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No.	Title	Authors	Year	Reason for Exclusion

Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
49	Financial Abnormal Data Detection System Based On Reinforcement Learning	Hou, X.	2022	Does not address re-search questions
50	A Forecasting Approach To Cryptocurrency Price Index Using Reinforcement Learning	Mariappan, L.T.; Pandian, A.; Kumar, V D.; Geman, O.; Chiuchisan, I.; Năstase, C.	2023	Insufficient Quality
51	Deep Equal Risk Pricing Of Financial Derivatives With NonTranslation Invariant Risk Measures †	Carbonneau, A.; Godin, F.	2023	Does not address research questions
52	Latent Segmentation Of Stock Trading Strategies Using Multi-Modal Imitation Learning	Maeda, I.; de-Graw, D.; Kitano, M.; Matsushima, H.; Izumi, K.; Sakaji, H.; Kato, A.	2020	Does not address research questions
53	Equal Risk Pricing Of Derivatives With Deep Hedging	Carbonneau, A.; Godin, F.	2021	Does not address research questions
54	I2Rl: Online Inverse Reinforcement Learning Under Occlusion	Arora, S.; Doshi, P.; Banerjee, B.	2021	Does not address research questions

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Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
55	On Parametric Optimal Execution And Machine Learning Surrogates	Chen, T.; Ludkovski, M.; Voß, M.	2023	Does not address research questions
56	A Generic Methodology For The Statistically Uniform & Comparable Evaluation Of Automated Trading Platform Components	Sokolovsky, A.; Arnaboldi, L.	2023	Does not address research questions
57	Curriculum Goal-Conditioned Imitation For Offline Reinforcement Learning	Feng, X.; Jiang, L.; Yu, X.; Xu, H.; Sun, X.; Wang, J.; Zhan, X.; Chan, W.K.	2022	Does not address research questions
58	Optimal Execution Of A Basket Of Stocks Under Regime-Switching Model Using Reinforcement Learning	Pemy, M.; Zhang, N.	2024	Full-text not available
59	Enhancing Profit From Stock Transactions Using Neural Networks	Roy Choudhury, A.; Abrishami, S.; Turek, M.; Kumar, P.	2020	Does not address research questions
60	On Testing Machine Learning Programs	Braiek, H.B.; Khomh, F.	2020	Does not address research questions

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Table C.6: List of Studies Excluded in Full-text Review and Reasons

No.	Title	Authors	Year	Reason for Exclusion
61	Machine Learning In Asset Management—Part 2: Portfolio Construction— Weight Optimization	Snow, D.	2020	Full-text not available
62	Optimal Attention Allocation: Picking Alpha Or Betting On Beta?	Gu, Z.; Shi, Y.; Yan, T.; Zhou, Y.	2024	Does not address research questions
63	Real-Time PricingBased Resource Allocation In Open Market Environments	Mishra, P.; Moustafa, A.; Ito, T.	2023	Does not address research questions
64	Deep Treasury Management For Banks	Englisch, H.; Krabichler, T.; Müller, K.J.; Schwarz, M.	2023	Does not address research questions
65	Wealth Flow Model: Online Portfolio Selection Based On Learning Wealth Flow Matrices	Yin, J.; Wang, R.; Guo, Y.; Bai, Y.; Ju, S.; Liu, W.; Huang, J.Z.	2022	Full-text not available
66	Dynamic Pricing Of Differentiated Products With Incomplete Information Based On Reinforcement Learning	Wang, C.; Cui, S.; Wu, R.; Wang, Z.	2022	Does not address research questions
67	Interactive Preference Analysis: A Reinforcement Learning Framework	Hu, X.; Kang, S.; Ren, L.; Zhu, S.	2024	Does not address research questions

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No. Title Authors Year Reason for Exclusion

Table C.6: List of Studies Excluded in Full-text Review and Reasons

68	Robo-Advising: Learning Investors' Risk Preferences Via Portfolio Choices	Alsabah, H.; Capponi, A.; Ruiz Lacedelli, O.; Stern, M.	2021	Full-text not available
69	Synergistic Integration Between Machine Learning And AgentBased Modeling: A Multidisciplinary Review	Zhang, W.; Valencia, A.; Chang, N.-B.	2023	Does not address research questions
70	Comparative Analysis Of Predictive Algorithms For Performance Measurement	Gupta, S.; Kishan, B.; Gulia, P.	2024	Does not address research questions
71	A Robust Real Time Bidding Strategy Against Inaccurate Ctr Predictions By Using Cluster Expected Win Rate	Shih, W.-Y.; Lai, H.-C.; Huang, J.-L.	2023	Does not address research questions
72	A Brief Review Of Portfolio Optimization Techniques	Gunjan, A.; Bhattacharyya, S.	2023	Does not address research questions
73	A Static-Dynamic Hypergraph Neural Network Framework Based On Residual Learning For Stock Recommen-dation	Hao, J.; Liu, Z.; Sun, Q.; Zhang, C.; Wang, J.	2024	Does not address research questions

Table C.6: List of Studies Excluded in Full-text Review and Reasons

Continued on next page

No.	Title	Authors	Year	Reason for Exclusion
74	Beating A Benchmark: Dynamic Programming May Not Be The Right Numerical Approach	Van Staden, P.M.; Forsyth, P.A.; Li, Y.	2023	Does not address research questions
75	Rlstm: A New Framework Of Stock Prediction By Using Random Noise For Overfitting Prevention	Zheng, H.; Zhou, Z.; Chen, J.	2021	Does not address research questions

Appendix D. Data Extraction Template

1 General Information

Question ID	Question
1	Study ID (Author Last Name, Year)
2	Title of the Paper
3	Journal Subject Area (Scimago classification)
4	Type of Study (Primary / Secondary)
5	What are the objectives of the study?
6	Which RL technique is used? (Policy-based, Actor-Critic, etc.)
7	Which finance application is tested? (Trading, Portfolio Mgmt., etc.)
8	What are the stated key results?
9	What are the stated limitations?

2 Dataset Details

Question ID	Question

10	What asset classes are in the dataset? (Stocks, Crypto, etc.)
11	Number of assets in the dataset
12	Nature of the dataset (Real-world, Synthetic, Mixed)
13	Granularity of price data (Daily, Minute, Millisecond)
14	Format of price data (Time Series, Order Book)
15	Other datasets used (Market, News, Social Media, etc.)
16	Timeframe of time series data (Start - End)
17	Is the dataset publicly available? (Yes/No)
18	Which continents are represented in the price dataset?
19	Is dataset quality discussed in the article?

Question ID	Question
24	Are all performance results published to avoid selection bias?
25	Is the process of hyperparameter tuning explicitly described?
26	Which evaluation metrics are used? (Sharpe Ratio, PnL, etc.)

Table E.13: Results of Visual Inspection of Code Repositories. *y* = yes, *n* = no, *?* = unclear.

27	Are results validated with statistical tests or confidence intervals?
28	Is the code or workflow publicly available?

3 Generalizability of Results

Question ID	Question
20	Is out-of-sample testing applied?
21	Was the model tested on diverse test sets?
22	Was the model tested across different datasets or applications?
23	Is the non-stationarity problem explicitly addressed?

4 Methodological Rigor

5 Practicality and Benchmarking

Question ID	Question
29	What is the implementation status? (Theoretical, Simulation, Real-world)
30	Does the study conduct a benchmark?
31	What is the model benchmarked against? (Market Index, Baselines, etc.)
32	Is a standard dataset used for all models?

6 Transfer Learning and Taxonomies

Question ID	Question
33	Does the study mention Transfer Learning?
34	Does the study mention alternative generalization methods?
35	Does the study propose a framework to structure RL research in finance?

Appendix E. Summary of Reproducibility Inspection

No.	Study ID	Comments	Rerun as is	New price data	Custom dataset
1	Carta21a	Input data not available. First data level is output of a preprocessing layer (predictions). Cannot test on new price data without rebuilding preprocessing.	y	n	n
2	Kochliaridis2023	Code supports rerunning and changing timeframes.	y	y	y
3	Lee2023	Fully rerunnable and timeframes adjustable.	y	y	y
4	Zengeler2020	No documentation. Uses 20+ assets with fixed splits; new data might work for some assets.	y	n	y
5	Marzban2023a	No documentation. Synthetic data only. Option pricing.	y	?	?
6	Kochliaridis2024	Large zip file. Minimal instructions. Partial data. Should allow dataset replacement.	y	n	y

Continued on next page

Table E.13: Results of Visual Inspection of Code Repositories. *y* = yes, *n* = no, *?* = unclear.

No.	Study ID	Comments	Rerun as is	New price data	Custom dataset
7	Chakole2021	Python files on Google Drive. Loads Yahoo/CSV data. Flexible dates. Lacks instructions. Mentions KMeans use.	y	n	y
8	Jing2024	One-file run example. Reads fixed CSV. Requires candle chart generation for new data.	y	n	y
9	Gasperov2022	Notebook file using synthetic environment. No real data. Unclear how to change for tests.	y	?	y
10	Giorgi2023	Raw XLSX files. Includes run.sh. Needs dataset creation for reproducibility.	y	n	y
11	Carta2021b	main.py runs experiment on fixed CSV data. New dataset creation needed for reproducibility.	y	n	y
12	Shi2021	Two environments with train/test files. Uses fixed CSV.	y	n	y

Continued on next page

No.	Study ID	Comments	Rerun as is	New price data	Custom dataset
13	Brini2023	Synthetic-only, unclear how to reproduce. Appears to be generic RL framework.	?	n	?
14	Marzban2023b	main.pyexecutable with CSV data.	y	n	y
15	Su2024	Repo mismatch: links to wrong paper or unrelated content. Potential double publication.	n	n	n
16	Liu2024b	Repo contains CSVs only. No code available.	n	n	n
17	Felizardo2022	One Python file. Data source needs in-code replacement.	y	n	y

Appendix F. RL in Finance Reproducibility Checklist

This checklist is adapted from the NeurIPS reproducibility checklist and aligned with the FAIR principles. It provides actionable guidance for ensuring reproducibility for RL in finance research.

Findable

1. Unique identifiers: Did you assign persistent identifiers to your code/data? *Example:* Zenodo DOI for a GitHub repo release; version tag in GitHub.

2. Rich metadata: Did you describe the dataset(s) and code with sufficient metadata? *Example:* “Daily equity data (2000–2020), adjusted for dividends and splits, 500 stocks, cleaned using script preprocess.py.”

Table E.13: Results of Visual Inspection of Code Repositories. *y* = yes, *n* = no, *?* = unclear.

3. Linkage: Does the metadata explicitly link to the dataset/code version used? *Example:* "Results in Section 4.2 use dataset v1.2 (DOI:10.5281/zenodo.12345)."
4. Indexed resource: Is your code/data indexed in a searchable resource? *Example:* Papers with Code entry, Hugging Face Datasets card, or GitHub repo link in the paper.

Accessible

1. Open access: Is the code/data retrievable via a standard, open protocol (HTTPS/Git)? *Example:* GitHub public repo, Hugging Face hub dataset.
2. Metadata availability: If data cannot be shared (e.g., WRDS license), is metadata and a preprocessing script still available? *Example:* "We cannot share CRSP, but preprocessing scripts and a synthetic equivalent dataset are provided."

Interoperable

1. Standard formats: Are data/code/configs in standard formats? *Example:* Data in CSV/Parquet; configs in YAML/JSON; experiments callable via Python API.
2. Community vocabularies: Do you use common naming conventions? *Example:* asset_class: equity, frequency: daily, exchange: NYSE instead of ad hoc variable names.
3. Cross-referencing: Do datasets, code, and configs explicitly reference each other? *Example:* Config file includes dataset DOI + preprocessing script name.

Reusable

1. Documentation: Is code documented with a README and comments explaining workflow? *Example:* Top-level README: installation steps, run instructions, description of files.

2. Licensing: Are code/data released under a clear license? *Example:* MIT/Apache 2.0 for code, CC-BY 4.0 for synthetic datasets.
3. Provenance: Is the origin and processing of data/code described? *Example:* "Original source: WRDS TAQ; cleaned with script clean_taq.py; resampled to 1min bars."
4. Domain standards: Are finance-specific reproducibility issues addressed? *Example:* Statement on survivorship bias, look-ahead bias, corporate actions, transaction costs.
5. One-run reproducibility: Can results be reproduced by running a single main script or notebook? *Example:* python run_all.py reproduces all tables and figures in the paper.

Appendix G. Formalization of Tasks

Prediction Task

Objective: Produce forecasts of financial variables (e.g., returns, volatility, directional movement of price) that can inform downstream allocation or execution tasks. Prediction itself does not generate profit and loss (PnL), but serves as an information-providing component.

Setup. Let (X,Y) denote the input and output spaces:

- Input features: $x_t \in \mathrm{X}$ (e.g., past prices, volumes, technical indicators, fundamentals).
- Target variable: $y_{t+1} \in \mathrm{Y}$ (e.g., next-period return, realized volatility, binary direction).

Prediction function. A prediction model $f_\theta : \mathrm{X} \to \mathrm{Y}$ maps current features x_t to a prediction $\hat{y}_t$:

$$\hat{y}_t = f_\theta(x_t), \qquad \hat{y}_t \in \mathrm{R}^k \tag{G.1}$$

with k being the number of dimensions of Y.

Objective function.. Performance is measured via a task-specific loss $\ell : \mathrm{Y} \times \mathrm{Y} \to \mathrm{R}_+$:

$$\mathrm{L}(\hat{y}_t, y_{t+1}) = \ell(\hat{y}_t, y_{t+1}), \qquad \text{(G.2)} \quad \ell \in \{\text{MSE, MAE, log-loss, cross-entropy, quantile loss,...}\}. \qquad \text{(G.3)}$$

Here, ℓ denotes the chosen loss function (e.g., MSE, MAE), and $\mathrm{L}(\hat{y}_t, y_{t+1})$ is the loss value computed at a specific prediction step t.

Examples.

- Return forecasting: $y_{t+1} = r_{t+1}$, with ℓ = MSE$(\hat{r}_t, r_{t+1})$.
- Directional prediction: $y_{t+1} \in \{-1, +1\}$, with ℓ = log-loss.
- Volatility prediction: $y_{t+1} = \sigma_{t+1}$, with ℓ = MAE$(\hat{\sigma}_t, \sigma_{t+1})$.

Evaluation. Typical evaluation metrics include out-of-sample R^2, hit rate (for directional forecasts) or AUC. Importantly, prediction quality is judged relative to economic baselines (e.g., random walk, historical mean).

Role in pipeline. The outputs $\hat{y}_t$ are not traded directly; they act as inputs to:

- Allocation tasks (e.g., portfolio weights w_t chosen based on $\hat{r}_t$).
- Execution tasks (e.g., order timing based on predicted volatility $\hat{\sigma}_t$).

Allocation Task

Objective: Decide how to allocate capital or exposures across assets or instruments to optimize a reward-risk trade-off. Allocation tasks directly determine portfolio outcomes and are central to trading, portfolio management, market making, and hedging.

Setup. Let $r_{t+1} \in \mathbb{R}^N$ denote the vector of asset returns between t and t+1, where N denotes the number of tradable assets.

- Decision variables: $w_t \in \mathcal{W} \subseteq \mathbb{R}^N$ are portfolio weights or positions at time t.
- Wealth: $V_t \in \mathbb{R}_+$ denotes portfolio value at time t.
- Frictions: $\text{cost}(w_t, w_{t-1}) = \gamma \, \|w_t - w_{t-1}\|_1$, where $\gamma > 0$ is the per-unit trading cost (e.g., bid-ask spread or market impact), and $\|w_t - w_{t-1}\|_1$ denotes the total portfolio turnover between $t - 1$ and t.

Wealth dynamics. The portfolio evolves according to:

$$V_{t+1} = V_t\Big(1 + w_t^\top r_{t+1} - \text{cost}(w_t, w_{t-1})\Big), \qquad \text{(G.4)}$$

where $\text{cost}(w_t, w_{t-1})$ is a unitless scalar denoting the proportional deduction from returns due to trading frictions.

Objective function. Allocation tasks involve solving an optimization problem at each time step t to determine the allocation $w_t \in \mathcal{W}$:

$$w_t^\star = \arg\max_{w_t \in \mathcal{W}} \big(\mathbb{E}[w_t^\top r_{t+1}] - \lambda\, \mathcal{R}(w_t) - \text{cost}(w_t, w_{t-1})\big), \qquad \text{(G.5)}$$

for tasks that trade off return, risk, and transaction cost.

In contrast, for risk management or hedging tasks, the objective may instead focus purely on risk minimization:

$$w_t^\star = \arg\min_{w_t \in \mathcal{W}} \mathcal{R}(w_t) \qquad \text{subject to structural constraints (e.g., beta neutrality).} \qquad \text{(G.6)}$$

where:

- W is the feasible set of allocations (e.g., budget, leverage, or regulatory constraints),
- $\mathrm{E}[w_t^\top r_{t+1}]$ is the expected return,
- $\mathrm{R}(w_t)$ is a risk measure (e.g., variance, CVaR), • cost(w_t,w_{t-1}) models

  transaction costs or frictions,
- and $\lambda \geq 0$ is the risk-aversion parameter.

Examples.

- Portfolio Management: choose weights w_t to maximize expected wealth growth or risk-adjusted return (e.g., via Sharpe ratio or Kelly criterion).
- Algorithmic Trading: express long/short positions dynamically through w_t to exploit predictive signals while controlling turnover and transaction costs.
- Market Making: select inventory positions w_t on the bid/ask side to balance execution flow and penalize inventory risk.
- Hedging and Risk Management: choose w_t to minimize a risk measure $\mathrm{R}(w_t)$ subject to structural constraints (e.g., hedge ratios to reduce variance, beta exposure, or tail risk).

Evaluation. Performance is assessed using reward-risk metrics such as:

- Return-based: CAGR, average excess return.
- Risk-adjusted: Sharpe ratio, Sortino ratio, information ratio.
- Risk-focused: volatility, CVaR, maximum drawdown.
- Stability: turnover, transaction cost impact.

Role in pipeline. Allocation tasks are typically downstream from prediction (signals $\hat{y}_t$ feed into position sizing), and upstream from execution (allocation decisions w_t generate orders that must be implemented efficiently).

Allocation Task

Objective. Decide how to allocate capital across N tradable assets to optimize a reward–risk trade-off. Allocation tasks directly determine portfolio outcomes and are central to trading, portfolio management, market making, and hedging.

Setup. Let $r_{t+1} \in \mathbb{R}^N$ denote the vector of *simple returns* of the N assets between t and $t + 1$.

- Asset returns: For each asset $i \in \{1, \dots, N\}$, let $P_t^{(i)}$ and $P_{t+1}^{(i)}$ denote its prices at times t and $t + 1$. The one–period *simple return* is

$$r_{t+1}^{(i)} = \frac{P_{t+1}^{(i)} - P_t^{(i)}}{P_t^{(i)}} = \frac{P_{t+1}^{(i)}}{P_t^{(i)}} - 1 .$$

  Collecting all assets gives the return vector

$$r_{t+1} = \left(r_{t+1}^{(1)}, \dots, r_{t+1}^{(N)}\right)^\top \in \mathbb{R}^N .$$

- Decision variables: $w_t \in W \subseteq \Delta^{N-1}$ are portfolio weights, where

$$\Delta^{N-1} := \left\{w \in \mathbb{R}^N : \mathbf{1}^\top w = 1,\ w_i \geq 0\right\},$$

  so each $w_t^{(i)}$ is the fraction of current wealth V_t invested in asset i.

- Wealth: $V_t \in \mathbb{R}_+$ denotes total portfolio value at time t.

- Frictions: proportional transaction costs

$$\text{cost}(w_t, w_{t-1}) = \gamma \, \|w_t - w_{t-1}\|_1,$$

where $\gamma > 0$ is the per-unit trading cost (e.g., bid–ask spread or market impact), and $\|w_t - w_{t-1}\|_1$ is total portfolio turnover between $t-1$ and t.

Wealth dynamics. Because w_t are fractions of wealth, the self-financing portfolio evolves as

$$V_{t+1} = V_t \left[1 + w_t^\top r_{t+1} - \mathrm{cost}(w_t, w_{t-1})\right],$$

where cost(·) is a unitless fraction representing the proportional deduction due to trading frictions.

Objective function.. At each time step t, the allocation is chosen by solving

$$w_t^\star = \arg\max_{w_t \in W} \left\{ \mathbb{E}_t[w_t^\top r_{t+1}] - \lambda R(w_t) - \mathrm{cost}(w_t, w_{t-1}) \right\},$$

for tasks that trade off expected return, risk, and transaction cost. For pure risk-management or hedging tasks, the objective may instead be

$$w_t^\star = \arg\min_{w_t \in W} R(w_t) \qquad \text{s.t. structural constraints (e.g., beta neutrality).}$$

Definitions

- W is the feasible set of allocations (e.g., leverage limits or regulatory constraints).
- $\mathbb{E}_t[w_t^\top r_{t+1}]$ is the conditional expected portfolio return given information at t.
- $R(w_t)$ is a risk measure (variance, CVaR, etc.).
- $\lambda \geq 0$ controls risk aversion.

Evaluation. Performance is assessed using reward–risk metrics such as:

- *Return-based:* CAGR, average excess return.

- *Risk-adjusted:* Sharpe ratio, Sortino ratio, information ratio.
- *Risk-focused:* volatility, CVaR, maximum drawdown.
- *Stability:* turnover, transaction-cost impact.

Role in pipeline. Allocation tasks are typically downstream from prediction (signals $\hat{y}_t$ feed into position sizing) and upstream from execution (the chosen weights w_t generate orders to be implemented efficiently).

Execution Task

Objective: Implement trades in financial markets while minimizing execution costs, market impact, and slippage. Execution tasks translate portfolio or trading decisions into concrete market orders.

Setup. Suppose a trader must execute an order of total size Q within a given time horizon T, across K trading venues.

- Inventory: q_t denotes the remaining unexecuted quantity of the order at time t (i.e., outstanding trade inventory), with $q_0 = Q$ and $q_T = 0$.
- Decision variables: $\mathbf{x}_t \in \mathbb{R}_+^K$ is the vector of order sizes placed across K venues at time t.
- Market state: $M_t \in \mathrm{M}^K$ is the vector of market states across venues at time t, with each $M_t^{(k)} \in \mathcal{M}$ capturing venue-specific features (spread, depth, volatility, etc.).

Inventory dynamics. The outstanding inventory evolves as:

$$q_{t+1} = q_t - \mathbf{1}^\top \mathbf{x}_t, \qquad q_0 = Q, \qquad q_T = 0, \tag{G.7}$$

where $\mathbf{1}^\top \mathbf{x}_t$ is the total executed size at time t across all venues.

Execution cost function. The realized execution cost at time t is:

$$C(\mathbf{x}_t, M_t) = \sum_{k=1}^{K} \Big((\text{PricePaid}_t^{(k)} - \text{Benchmark}_t) x_t^{(k)} + \text{impact}(x_t^{(k)}, M_t^{(k)}) \Big) + \kappa q_t^2, \tag{G.8}$$

where:

- $\text{PricePaid}_t^{(k)}$ is the execution price at venue k and time t,
- Benchmark_t is the reference price (e.g., arrival, VWAP, TWAP),
- $\text{impact}(x_t^{(k)}, M_t^{(k)})$ models temporary or permanent market impact at venue k,
- κq_t^2 penalizes residual inventory risk at time t.

Objective function. The execution problem seeks to minimize the expected cumulative cost of trading, subject to completing the total order:

$$\min_{\{\mathbf{x}_t\}_{t=0}^{T}} \mathbb{E}\left[\sum_{t=0}^{T} C(\mathbf{x}_t, M_t)\right] \quad \text{s.t.} \sum_{t=0}^{T} \mathbf{1}^\top \mathbf{x}_t = Q. \tag{G.9}$$

Cases.

- Optimal Order Execution (OEE): Single-venue case. The decision reduces to choosing scalar trade sizes $x_t \in \mathbb{R}_+$ (as $K = 1$) over time to execute the total order Q, with inventory dynamics $q_{t+1} = q_t - x_t$.
- Smart Order Routing (SOR): Multi-venue case. The decision variable becomes a vector $\mathbf{x}_t \in \mathbb{R}_+^K$, where $x_t^{(k)}$ denotes the order size at venue k at time t. The trader optimally allocates execution across both time and venues, based on market state $M_t = (M_t^{(1)}, \dots, M_t^{(K)})$, to minimize cost while satisfying $\sum_{t=0}^{T} \mathbf{1}^\top \mathbf{x}_t = Q$.

Evaluation. Execution quality is typically benchmarked using:

- Implementation Shortfall (difference between executed price and arrival price).
- VWAP/TWAP slippage.
- Cost relative to simulated optimal execution schedules (e.g., Almgren–Chriss model).
- Fill ratios and latency impact in high-frequency contexts.

Role in pipeline. Execution tasks consume orders generated by allocation tasks and convert them into actual trades. While prediction and allocation determine *what* to trade, execution determines *how* to trade under market frictions.

Appendix H. Data Card (Template)

The Data Card serves as a minimum reporting requirement for datasets used for RL in finance. It provides baseline transparency and comparability across studies. Researchers may extend this template to cover additional modalities (e.g., textual, order book, or alternative data), but at a minimum the following fields should be reported.[9]

Overview

Field	Value
Entities	[e.g. number of tickers]
Variables	[e.g. OHLCV, fundamental data]
Observations	[number of observations (time steps)]
Time range	[earliest timestamp - latest timestamp]
Inferred frequency	[e.g. minute, daily]

[9] [removed for anonymity]

Data Quality

Check	Value
Missing values	[ ]
Duplicates	[ ]
Timestamps monotonic per entity	[✓/]

Variables (All Entities)

Variable	Obs	Missing	Min	Max
Var1	[ ]	[ ]	[ ]	[ ]
Var2	[ ]	[ ]	[ ]	[ ]
...				

Per-Entity Variable Statistics Entity: [Name]

Field	Value
Entity observations	[ ]
Entity variables	[ ]
Entity time range	[ ]

Variable	Obs	Missing	Min	Max
Var1	[ ]	[ ]	[ ]	[ ]
Var2	[ ]	[ ]	[ ]	[ ]
...				

(Repeat per entity as needed.)

Appendix I. Model Card (Template)

The Model Card serves as a minimum reporting requirement for RL models in finance. It provides baseline transparency about model specification, assumptions, and intended use. Researchers may extend this template with additional details, but at a minimum the following fields should be reported.

Core (all models)

Field	Value
Model name / identifier	[ ]
Authors / institution	[ ]
Model type	[Supervised / RL / Heuristic / Other]
Task type	[Prediction / Allocation / Execution]
Intended use	[Research prototype / educational / production candidate]
Not intended use	[Financial advice / live deployment without validation]
Training setup	[Splits, hyperparameters, compute resources]
Limitations & risks	[Known weaknesses, assumptions, failure modes]

Task-specific extensions Prediction

Field	Value
Target variable	[Return, volatility, order flow, regime label]
Forecast horizon	[1d, 5min, etc.]
Model objective	[MSE, cross-entropy, custom loss]
Output type	[Continuous regression / classification probabilities]

llocation

Field	Value
Action space	[Portfolio weights, buy/sell/hold]

Constraints	[Position limits, leverage, turnover constraints]
Reward function	[PnL, log returns, risk-adjusted objective]
Rebalancing frequency	[Daily, intraday, event-driven]

Execution

Field	Value
Action space	[Order types: market, limit, child orders]
Latency model	[Assumed delay / simulation setup]
Impact model	[Almgren-Chriss/ decay/ LOB microstructure]
Risk & inventory controls	[Inventory limits, exposure constraints]
Reward function	[Implementation shortfall, cost minimization, spread capture]

Appendix J. Evaluation Card (Template)

The Evaluation Card serves as a minimum reporting requirement for documenting how models are assessed for RL in finance. It ensures comparability across studies by requiring core robustness checks and task-specific evaluation metrics. All reported metrics must include the mean, standard deviation, and a 95% confidence interval computed across all random seeds.

General Information

Field	Value
Task Type	[Allocation / Prediction / Execution]
Evaluation Periods	[Train, validation, test splits; multiple test windows if used]
Random Seeds	[Number of seeds, how aggregated]

Prediction — Regression

Metric	Value (mean ± std and 95% CI)

MSE / RMSE	[ ]
MAE	[ ]
$R2$	[ ]

Prediction — Classification

Metric	Value (mean ± std and 95% CI)
Accuracy	[ ]
Precision	[ ]
Recall	[ ]
F1 Score	[ ]
AUC-ROC	[ ]

Allocation

Metric	Value (mean ± std and 95% CI)
Annualized Return (CAGR)	[ ]
Volatility (annualized std. dev.)	[ ]
Maximum Drawdown (MDD)	[ ]
Sharpe Ratio	[ ]
Turnover	[ ]
Transaction Cost Assumptions	[e.g., 5 bps per trade]
Net vs. Gross Performance (transaction costs)	[ ]

Execution

Metric	Value (mean ± std and 95% CI)
Implementation Shortfall	[ ]
VWAP/TWAP Slippage	[ ]
Price Impact per Unit Traded	[ ]
Variance of Execution Costs	[ ]

Tail Risk of Execution Costs (e.g., 95th percentile)	[ ]